

YOUNES BOUCHBOUK

Software Engineer — Go/React Specialist — Building Scalable Cloud-Native Systems

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Experience

OmniLab by 21Factory

Sep 2024 – Present

Software Engineer (Full-Stack → Backend)

Casablanca, Morocco

- Joined the company during its early scaling phase and played a key role in evolving the platform from a small-scale architecture into a distributed ecosystem serving **millions of users** and high-throughput transactional workloads.
- Designed, architected, and maintained multiple production-grade microservices across **Booking, Games, and Smartlinks** domains using **Domain-Driven Design (DDD)** and **Hexagonal Architecture**.
- Led the transition toward a scalable **Event-Driven Architecture (EDA)** by implementing asynchronous workflows with the **Outbox Pattern, Debezium CDC, and NATS JetStream**, significantly improving system decoupling, resiliency, and scalability.
- Built and optimized distributed communication layers using **gRPC** and internal platform standards inspired by **Google AIP**, enabling reliable and high-performance inter-service communication.
- Developed internal observability and performance engineering solutions using **OpenTelemetry, Prometheus, and Datadog**, including custom dashboards, tracing pipelines, and load-testing analysis tools to proactively identify bottlenecks and optimize production systems.
- Improved platform reliability and engineering velocity through automated **integration testing, TDD practices**, transactional consistency strategies, query optimization, production debugging, and Kubernetes-based infrastructure operations.
- Contributed across backend systems, cloud infrastructure, CI/CD pipelines, and React/Next.js frontend applications in a fast-paced startup environment requiring strong ownership, rapid iteration, and cross-functional problem solving.
- Tech Stack:** Go, gRPC, NATS JetStream, Debezium, Kubernetes, Docker, MongoDB, ClickHouse, OpenTelemetry, Prometheus, Datadog, React.js, Next.js, GitLab CI/CD, TDD

Oracle Corporation

Jan – Jun 2024

Research Assistant (Back-End), Internship

Casablanca, MA

- Improved and automated testing processes for Graal Cloud Native Guides (graal.cloud/gcn/guides).
- Developed new features and fixed bugs for a notebook-based data science platform (Oracle Labs Data Studio).
- Worked on Jupyter integration: themes, system file access, and embedding Jupyter via iframes.
- Tech Stack::** Javascript, TypeScript, Python, Jupyter Lab, Docker, Kubernetes.

ISPApp

Aug 2022 – Jul 2023

Full-Stack Developer

Dover, USA (Remote)

- Developed a user-centric dashboard that streamlined router management for ISPs.
- Integrated **Plotly** for real-time visualization of network metrics (ping, TCP, interfaces, wireless APs).
- Tech Stack:** React.js, Next.js, Golang, Node.js, TailwindCSS, Plotly.

Projects

JOYO: Mental Health & Self-Awareness Platform | AI, Go, Sqlc, Docker, PostgreSQL, OpenAI API, DDD, GCP

- Spearheaded the backend development for the public launch, implementing core features driven by Domain-Driven Design (DDD) principles.
- Architected an **AI-driven self-exploration engine** using LLM integrations to generate personalized user prompts based on historical engagement data.
- Optimized the development lifecycle by implementing TDD, hot-reloading environments, and reducing Docker image sizes by 45% for faster CI/CD deployments.
- Enhanced system performance and reduced API latency by implementing efficient caching strategies and database query optimization.

ResumeForge AI (Microservices Monorepo) | Go, Node.js, Next.js, gRPC, MongoDB, Redis

- Engineered a distributed AI-powered resume system using a monorepo architecture, featuring dedicated services for Auth, Profile Management, and AI Matching.

- Implemented secure inter-service communication using gRPC for internal binary calls and REST for external client-facing APIs.

SimpleBank: High-Consistency Financial Backend | *Go, PostgreSQL, Kubernetes, AWS, gRPC*

- Built a transactional banking system in Go, implementing **DB isolation levels** and ACID-compliant transfers to ensure zero data loss during concurrent operations.
- Designed a dual-protocol server handling gRPC and HTTP/REST simultaneously via gRPC-gateway, documented with integrated Swagger UI.
- Deployed to **AWS EKS (Kubernetes)** with a fully automated **CI/CD pipeline** using GitHub Actions, Docker, and AWS ECR.

SENDIT: Full-Stack Delivery Network Application | *React, Node.js, MongoDB, Neo4j*

- Architected a logistics platform using the MERN stack integrated with **Neo4j Graph Database** for shortest-path route optimization between 50+ Moroccan cities.
- Supports multi-hop deliveries across disparate logistics providers, optimizing for both speed and cost.

Technical Skills

Programming Languages: Golang, JavaScript, TypeScript, Java, Python, Bash.

Infra & Backend & Architecture & Practices: Microservices, Agentic Engineering, Event Driven Architecture, Domain-Driven Design, Hexagonal/Clean Architecture, gRPC, Google AIP, REST APIs, Caching, QA testing, Atomic counters, TDD, OLAP, NATs Jetstream, Debezium, Outbox Pattern, AWS

Observability & Monitoring: Datadog, OpenTelemetry, ClickStack, ClickHouse, Prometheus.

DevOps & DevSecOps & Cloud: Docker, Kubernetes, GitLab CI/CD, SAST, DAST, SCA, Trivy, Snyk, AWS.

Frontend: React.js, Next.js, Three.js, TailwindCSS.

Databases: PostgreSQL, MongoDB, MySQL, Redis.

Tools: Git, Cursor, Claude Code, JIRA, Confluence, Linux.

Education

National Institute Of Posts And Telecommunications (INPT) <i>Engineer Degree: Data and Software Engineering</i>	Sep 2022 – Jul 2024 <i>Rabat, MA</i>
Polydisciplinary Faculty of Taroudant (FP) <i>Professional License: Computer Science</i>	Sep 2021 – Jul 2022 <i>Taroudant, MA</i>
Superior School of Technology (EST) <i>Technical University Degree: Computer Science</i>	Sep 2019 – Jul 2021 <i>Safi, MA</i>